

HW 1. Solution:

T1: (1) 极坐标下 $\Delta_2 u = \frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0$

由于 $u = u(r)$ 故 $\Delta_2 u = \frac{d^2 u}{dr^2} + \frac{1}{r} \frac{du}{dr} = 0 \quad (r \neq 0)$

$\Rightarrow r^2 \frac{d^2 u}{dr^2} + r \frac{du}{dr} = 0$ 该方程为欧拉方程

作变量替换 $r = e^t$ 则有 $r \frac{du}{dr} = \frac{du}{dt}$ ($\frac{du}{dr} = \frac{du}{dt} \frac{dt}{dr} = \frac{1}{r} \frac{du}{dt}$)

且 $\frac{d^2 u}{dr^2} = \frac{d(\frac{1}{r} \frac{du}{dt})}{dr} = -\frac{1}{r^2} \frac{du}{dt} + \frac{1}{r} \frac{d(\frac{du}{dt})}{dr}$
 $= -\frac{1}{r^2} \frac{du}{dt} + \frac{1}{r} \frac{d(\frac{du}{dt})}{dt} \frac{dt}{dr} = -\frac{1}{r^2} \frac{du}{dt} + \frac{1}{r^2} \frac{d^2 u}{dt^2}$

即: $r^2 \frac{d^2 u}{dr^2} = -\frac{du}{dt} + \frac{d^2 u}{dt^2}$ 故原式化为 $\frac{d^2 u}{dt^2} = 0$

故通解为: $u(t) = C_1 + C_2 t \Rightarrow u(r) = C_1 + C_2 \ln r \quad (r > 0)$

(2)

$$\Delta_3 u = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 \frac{\partial u}{\partial r}) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \frac{\partial u}{\partial \theta}) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 u}{\partial \varphi^2}$$

由于 $u = u(r)$ 故 $\Delta_3 u = \frac{2}{r} \frac{du}{dr} + \frac{d^2 u}{dr^2}$

故 $\Delta_3 u + k^2 u = \frac{d^2 u}{dr^2} + \frac{2}{r} \frac{du}{dr} + k^2 u = 0$

作变量代换有: $v(r) = ru(r)$ $\Rightarrow v'' + k^2 v = 0$

为了消除一阶导数项
特征方程为: $\lambda^2 + k^2 = 0 \Rightarrow \lambda = \pm ik$

故 $v(r) = C_1 \cos kr + C_2 \sin kr$, C_1, C_2 为常数

$$u(r) = \frac{v(r)}{r} = \frac{C_1 \cos kr + C_2 \sin kr}{r} \quad (r > 0)$$

欧拉方程: $x^n y^{(n)} + a_{n-1} x^{n-1} y^{(n-1)} + \dots + a_1 x y' + a_0 y = 0$

例: 二阶欧拉方程: $x^2 y'' + b x y' + c y = 0$ (1) [其中 $y = f(x)$]

解法: ① 令 $x = e^t$ 代入 (1) 式得: $\frac{d^2 y}{dt^2} + (b-1) \frac{dy}{dt} + c y = 0$ 设 $y = e^{kt}$ 有

特征方程 $k^2 + (b-1)k + c = 0$ 解出 k_1, k_2 依据 k_1, k_2 分为三种情况:

$$\begin{cases} \text{实根且不同} & y = c_1 e^{k_1 t} + c_2 e^{k_2 t} = c_1 x^{k_1} + c_2 x^{k_2} \\ \text{实根且重根} & y = (c_1 + c_2 t) e^{kt} = (c_1 + c_2 \ln x) x^k \quad (k_1 = k_2 = k) \\ \text{复根} & k = \alpha \pm i\beta \text{ 则 } y = x^\alpha [c_1 \cos(\beta \ln x) + c_2 \sin(\beta \ln x)] \end{cases}$$

② 假设 $y = x^k$ 代入(1)式得: $(k^2 + (b-1)k + c) x^k = 0$

故 $k^2 + (b-1)k + c = 0$ 解 k_1, k_2 同上.

T2.

$$u_x = F' + G' \quad u_{xx} = F'' + G''$$

$$u_y = F' \lambda_1 + G' \lambda_2 \quad u_{yy} = F'' \lambda_1^2 + G'' \lambda_2^2$$

$$u_{xy} = F'' \lambda_1 + G'' \lambda_2$$

代入原式: $u_{yy} - (\lambda_1 + \lambda_2) u_{xy} + \lambda_1 \lambda_2 u_{xx}$

得: 该式 = 0

T3:

$$u_t = -\frac{1}{2} t^{-\frac{3}{2}} \exp\left\{-\frac{(x-\xi)^2}{4a^2 t}\right\} + \frac{1}{\sqrt{t}} \cdot \exp\left\{-\frac{(x-\xi)^2}{4a^2 t}\right\}.$$

$$= \left(-\frac{1}{2t} + \frac{(x-\xi)^2}{4a^2 t^2}\right) \frac{1}{\sqrt{t}} \exp\left\{-\frac{(x-\xi)^2}{4a^2 t}\right\} \quad \frac{(x-\xi)^2}{4a^2} t^{-2}$$

$$u_x = \frac{1}{\sqrt{t}} \exp\left\{-\frac{(x-\xi)^2}{4a^2 t}\right\} \cdot \left(-\frac{x-\xi}{2a^2 t}\right)$$

$$\text{故 } u_{xx} = \left(\frac{(x-\xi)^2}{4a^4 t^2} - \frac{1}{2a^2 t}\right) \frac{1}{\sqrt{t}} \exp\left\{-\frac{(x-\xi)^2}{4a^2 t}\right\}$$

$$\text{故 } u_t = a^2 u_{xx}$$

令 $\frac{1}{t} = s$, 当 $t \rightarrow 0^+$ 时, $s \rightarrow +\infty$

$$u = \frac{\sqrt{s}}{e^{\frac{(x-\xi)^2}{4a^2} s}} \quad \text{由于 } x \neq \xi \text{ 记 } \frac{(x-\xi)^2}{4a^2} = m > 0$$

故 $u = \frac{\sqrt{s}}{e^{ms}}$ 由于指数函数增长速度远快于幂函数

$$\text{则 } \lim_{t \rightarrow 0} u(t, x) = \lim_{s \rightarrow +\infty} u = 0 \quad (x \neq \frac{1}{2})$$

T4.

$$u_x = a e^{2x+y} + a x e^{2x+y}, 2 = (a + 2ax) e^{2x+y}$$

$$u_{xx} = 2a e^{2x+y} + (a + 2ax) e^{2x+y} \cdot 2 = (4a + 4ax) e^{2x+y}$$

$$u_y = a x e^{2x+y} \quad u_{yy} = a x e^{2x+y}$$

代入方程中得: $4a e^{2x+y} = e^{2x+y}$ 由于 $e^{2x+y} > 0$

故 $4a = 1$ 即 $a = \frac{1}{4}$ 故特解为 $u = \frac{1}{4} x e^{2x+y}$

T5:

$$u_x = f' y \quad u_y = f' x$$

$$x u_x - y u_y = x y f' - x y f' = 0 \quad \text{故得证}$$